

REVA AGRAWAL

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EDUCATION

University of California San Diego

Bachelor of Science in Data Science with a minor in Mathematics | GPA 3.7

Sept 2022 - June 2026

- **Coursework:** Machine Learning Algorithms, Applied Generative AI, Recommender Systems, Database Management & Query Optimization, Data Visualization, Advanced Data Structures, Business Intelligence & Analytics, Probability & Statistics
- **Research:** Migrated MATLAB code to Python for dimensionality estimation, time-dependent decoding, and dynamic neural visualization in a neuroscience study on Pre-Frontal Cortex dynamics (Aoi et al., bioRxiv 2019) under Dr. Mikio Aoi.

WORK EXPERIENCE

Data Scientist

IDX Exchange, Baltimore, Maryland

Sept 2025 - Dec 2025

- Built a Python ETL and data quality workflow with pandas and NumPy to process **200K+ California property records**, filtered to single family homes, and produced model ready datasets with imputation and categorical encoding.
- Trained and fine tuned XGBoost, CatBoost, and Random Forest regressors with Optuna optimization and a train validation test setup, achieving approximately **12% Mean Absolute Percentage Error** on held out test data.
- Engineered **30+ temporal, ratio, and geographic features** to improve signal and stability, validated via k fold cross validation with residual diagnostics and segment level error analysis.

Fullstack Software Engineer - Data & AI

Gallows AI, Tokyo, Japan

June 2025 - Aug 2025

- Engineered Python backend repositories and controller, router, and service layers (FastAPI, SQLAlchemy) to support CRUD operations with dynamic query filtering for **5K+ customer interactions**.
- Standardized schemas, migrations, and setup scripts (Pydantic, Alembic, PostgreSQL), reducing developer onboarding time and preventing integration issues on multi-million row datasets.
- Implemented WebSocket infrastructure for Twilio and reduced **call response latency from ~3 seconds to 0.5 seconds**.

Software Development Engineer

IEEE-HKN, Kappa Psi Chapter, San Diego

Feb 2025 - Present

- Led 8 developers to build the frontend architecture for a modular 9-key programmable macro pad, utilizing QML and C++.
- Reduced macro pad **setup time from 20 minutes to 6 minutes** by delivering a cross-platform UI (Python + PyQt) with clear error handling and logging, enabling non-technical users to configure custom macros.
- Designed and implemented dynamic image rendering for keycaps to enable **50+ custom configurations per device**.

Machine Learning Intern

Spartificial, Bangalore, India

June 2024 - Sept 2024

- Developed a hybrid deep learning pipeline using CNN and U-Net architectures for semantic segmentation of multispectral satellite imagery, achieving a **14% improvement in IoU** on 12+ underrepresented land classes.
- Optimized model inference through quantization-aware training, streamlining data preprocessing overhead from **4.0s to ~1.8s per batch of 64 images** via parallelized NumPy operations and memory efficient data loaders.
- Conducted targeted hyperparameter tuning to increase reliability in low-sample regions; deployed models via TensorFlow.

Project Engineer

Triton Neurotech, San Diego

Dec 2023 - Apr 2024

- Developed a myoelectric prosthetic arm achieving **85%+ gesture classification accuracy** with EMG signals from OpenBCI Cyton boards, with testing and validation pipelines for signal denoising (Butterworth filters, RMS feature extraction).
- Trained SVM and k-NN models and prototyped 1D CNN and LSTM time-series models to evaluate classification fidelity.

PROJECTS

Budget Leakage Detection Lakehouse | PySpark, SQL, Parquet, ETL, AWS S3, Snowflake, Airflow, Streamlit

- Built a PySpark data lakehouse processing \$6.7M in expenses (1,500+ transactions) and detecting 2,263+ leakage signals via 5 fraud algorithms; designed Parquet-based star schema for analytics.
- Developed Airflow-orchestrated ETL with data quality checks; deployed Streamlit dashboards and SQL BI views to budget variances, vendor concentration risk, and leakage patterns across 50+ vendors and 84 budget records.

Differentially Private Telemetry Analysis | Python, NumPy, Pandas, Scipy, Statistical Analysis

- Built an analytics system processing 152K+ telemetry events across 137K+ users using Gaussian and Laplace differential privacy mechanisms to accurately identify high-error products while protecting individual user privacy.

Img2LaTeX AI | Python, FastAPI, React, Qwen2-VL, PyTorch, SQLite

- Recognized at DiamondHacks 2025 for building an AI platform converting math equation images to LaTeX using Hugging Face Transformers and Qwen2-VL models with GPU-accelerated inference and dynamic model selection.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, SQL, Java, JavaScript, HTML/CSS, React, Next.js, Node.js, FastAPI, C++
- **ML & Data Libraries:** Scikit-Learn, TensorFlow, NumPy, XGBoost, Pandas, NLTK, U-Net, CNN, LSTM, PyTorch
- **Cloud & DevOps:** Docker, Git, CI/CD pipelines (GitHub Actions, GitLab CI), AWS (S3, EC2), Azure
- **Databases & Warehousing:** PostgreSQL, MySQL, MongoDB, Snowflake, Redis, SQLAlchemy, Alembic
- **Data Engineering:** Apache Spark, ETL/ELT Pipelines, Data Modeling, Workflow Orchestration
- **Data Visualization & BI:** Tableau, Power BI, Matplotlib, Seaborn, Plotly, D3.js
- **Core Competencies:** Linux/Unix scripting, Data Wrangling, Statistical Modeling, A/B Testing, Agile